

VELMÈRE BASIC AUDIT REPORT

Polkadot Relay Chain (native-polkadot-relay)

VELMERE SECURITY AUDIT REPORT: POLKADOT RELAY CHAIN

Audited Contract: native-polkadot-relay
Network: Polkadot Relay Chain (Chain ID: 392)
Report ID: vlm_dot_basic_365 | Tier: BASIC | Application Surface: BROWSER
Created At: 2026-09-07T21:38:33.396Z

VERDICT SUMMARY: NOT SCORED (MISSING BYTECODE)
Confidence Score: 0/100 | Evidence Coverage: 0%
Bytecode is missing or malformed. In accordance with Velmère truth invariants, bytecode-derived claims are NOT emitted and security risk is marked NOT SCORED.

--- PROTOCOL OVERVIEW & NATIVE L1 NETWORK CONTEXT [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Summary of distributed architecture and consensus mechanism
Protocol / Asset: Polkadot Relay Chain
Architecture Type: Native Base Layer (Layer-1 · No Smart Contract)
Network Identifier: native_chain:dot
Native Symbol: DOT
Rule Governance: Distributed node consensus across validators / miners

--- CONSENSUS SPECIFICATION & NODE CLIENT VERIFICATION [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Analysis of block validation rules, transactions, and cryptographic curves
Verification of Polkadot Relay Chain is conducted against native Layer-1 consensus specifications. EVM bytecode decompilation is NOT_APPLICABLE for native base protocol assets.

--- BASELINE NETWORK SECURITY FINDINGS [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Identified risk vectors in base layer architecture

--- CONSENSUS GOVERNANCE & PROTOCOL UPGRADE ANALYSIS (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Decentralization evaluation, validator distribution, and improvement proposal process
[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Distributed consensus governance and protocol upgrade mechanism analysis
 - * Validator / miner distribution and network finality threshold evaluation
 - * Hard-fork coordination history and client implementation diversity audit
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- SPOT MARKET DEPTH, EXCHANGE RESERVES & NETWORK VELOCITY (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Global market liquidity, supply velocity, and mempool dynamics
[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Global order book depth across regulated spot venues and custodial reserves
- * On-chain circulating supply velocity and active address distribution
- * Network transaction fee volatility and mempool congestion behavior
- * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.

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--- NETWORK ATTACK VECTORS & CONSENSUS RESILIENCE (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Modeling of chain reorganizations, Sybil attacks, and P2P partitioning

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Consensus reorganization and majority hashrate/stake attack modeling
 - * P2P network partition, eclipse attack, and sybil resistance analysis
 - * Mempool front-running, transaction censorship, and MEV extraction vectors
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
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--- STATE ENGINE DETERMINISM & MULTI-CLIENT WIRE COMPATIBILITY (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Binary compatibility verification and state transition engine determinism

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Reference node client reproducible build verification against repository tags
 - * Consensus serialization and wire protocol binary compatibility analysis
 - * State transition engine deterministic execution boundary checks
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- PROTOCOL HARD/SOFT FORK EVOLUTION & INVARIANT PROOFS (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Tracking of consensus rule evolution and backward compatibility guarantees

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Protocol upgrade proposal activation and deployment history
 - * Backward compatibility verification of block validation logic
 - * Formal verification: NOT EXECUTED (Formal verification engine not commissioned)
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
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--- HUMAN ANALYST REVIEW & PROTOCOL ARCHITECTURE AUDIT (ADVANCED) [ADVANCED] ---

VERIFICATION TYPE: INDEPENDENT MANUAL AUDITOR REVIEW

Attestation by independent distributed systems auditor

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Analyst sign-off requires verified reviewer evidence
 - * Cryptographically signed analyst attestation digest
 - * Manual review boundary explicit: no guarantee of absolute perfection
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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